
# Electrical Measurement

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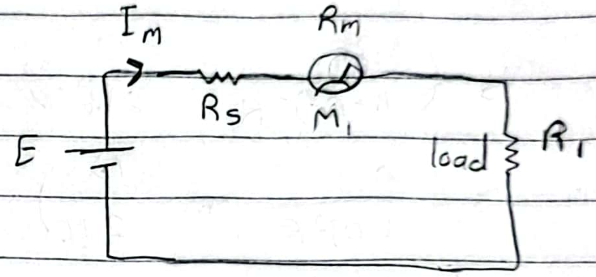
# Current meters (Ammeter)

rules for using current meters:-

1- Connect series with load
2- ~~Mu~~ should full scale current greater that max current
3- internal resistance should be smaller than load by 10 times

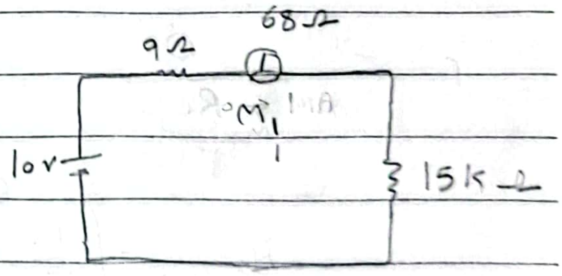
$$I_m = \frac{E}{R_s + R_m + R_1}$$



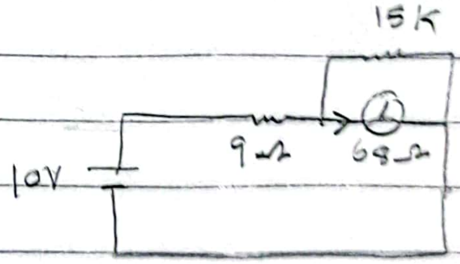

ex:- Find the current in $M_1$

$$I_m = \frac{E}{R_s + R_m + R_1} = \frac{10}{9 + 68 + 15k}$$

$= 0.67\ mA$




ex:- find the current

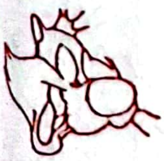



~~For~~ $R_1 \gg R_m$

$R_{total} = R_m + R_s = 68 + 9$

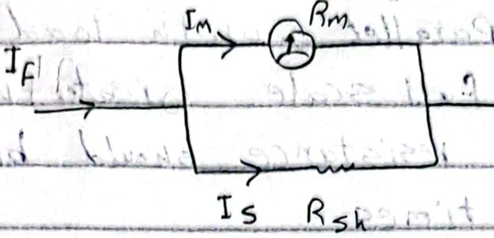
$= 77\ \Omega$

$$I_m = I_{total} = \frac{E}{R_{total}} = \frac{10}{77} = 130\ mA$$



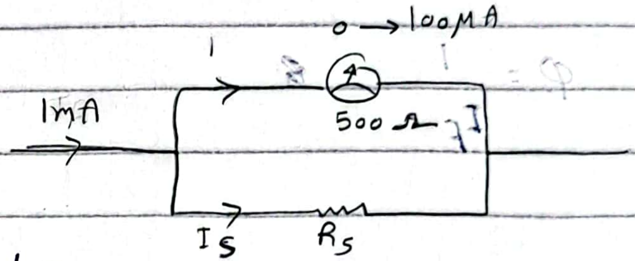
so to get current meter with high full scale
We should add ~~shun~~ resistance in Parallel with M

$$I_m = I_f\left(\frac{R_s}{R_m+R_{sh}}\right)$$




$$I_f = I_m + I_s$$

$$R_{sh} = \frac{E_m}{I_s} = \frac{I_m R_m}{I_f - I_m}$$



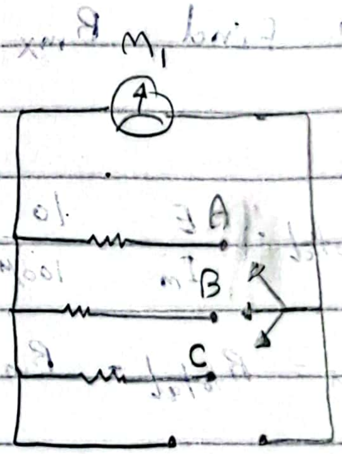

ex: find $I_s$ and $R_s$

$$I_s = I_f - I_m = 1m - 0.1m$$
$$= 0.9\,mA$$

$$R_s = \frac{E_m}{I_s} = \frac{I_m R_m}{I_s} = \frac{500 \times 100\mu}{0.9\,m} = 55.6\,\Omega$$

multiple -Range ammeter

| | | full-scale |
|---|---|---|
| A | → | 1mA |
| B | → | 10mA |
| C | → | 100mA |



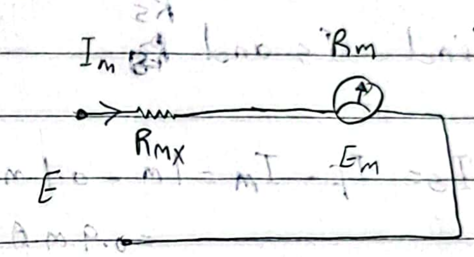
# Voltmeter

Rules for using voltmeter

1- Connect parallel with load
2- should full scale greater than max voltage
3- internal resistance should be greater than load by 100 times

Voltmeter sensitivity ($\phi$)

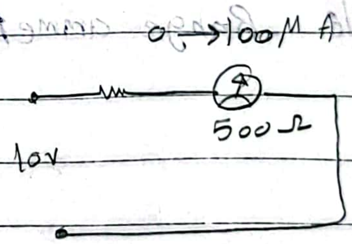
$$\phi = \frac{1}{I_F}$$




$$I_m = \frac{E}{R_m + R_{mx}}$$

$$R_m = \phi \times E \qquad E_m = E\left(\frac{R_m}{R_m + R_{mx}}\right)$$

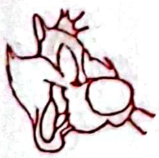
ex 1- find $R_{mx}$ and $E_m$




$$R_{total} = \frac{E}{I_m} = \frac{10}{100\mu} = 100\,000\ \Omega$$

$$R_{mx} = R_{total} - R_m = 100\,000 - 500 = 99\,500\ \Omega$$

$$E_m = I_m R_m = 100\mu \times 500 = 0.05\ V$$



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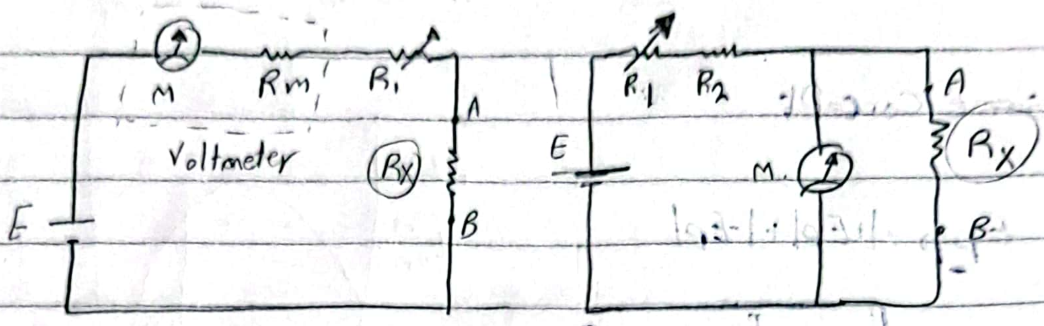
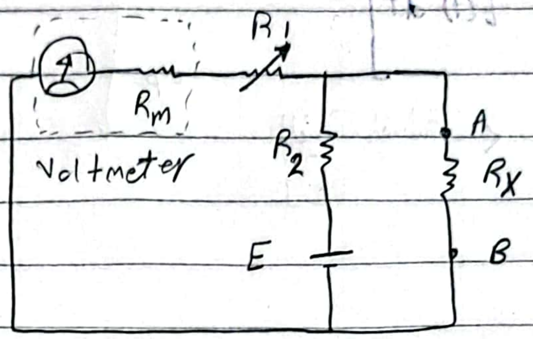
ohmeter:




Rx : resistance that needed to determine value

multimeter:

instrument is used to measure , current, voltage and resistance

ohmeter + voltmeter + ammeter

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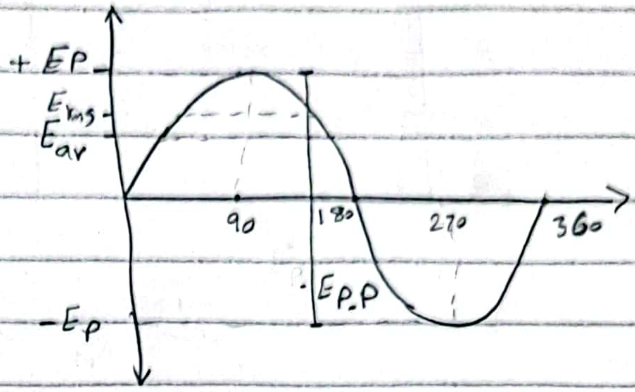
# AC Current

Some Concept

$$E_{P\text{-}P} = |+E_P| + |-E_P|$$




$$E_{rms} = \sqrt{\left[\frac{1}{T}\int_0^T E(t)\,dt\right]^2}$$

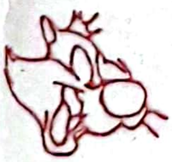
$$= \frac{E_P}{\sqrt{2}} \leftarrow$$ القيمة الفعلية لـ AC

$$E_{av} = \frac{2E_P}{\pi}$$

Form Factor (A) $= \dfrac{E_{rms}}{V_{av}} = \dfrac{0.7\,E_P}{0.6\,E_P} = 1.1$

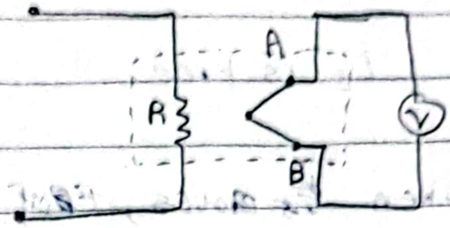
## AC meter

1- thermocouple AC meter
2- Hot wire Ammeter
3- electrodynamometer
4- Iron-~~Vane~~ Vane meter



Thermo Couple

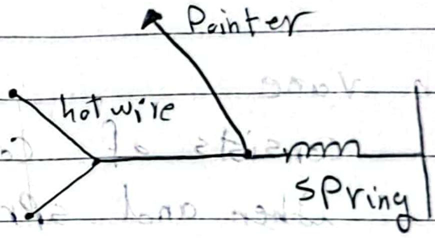
depend ~~[illegible]~~ on thermoCouple sensor that measure heat from AC current then convert to voltage that useable




measure r.m.s value

Hot-wire

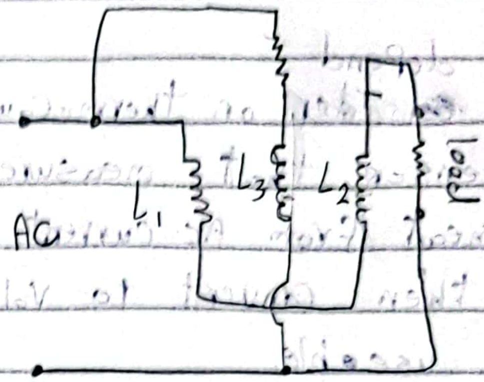
when ~~[illegible]~~ current flow in hot wire, it expands and Pointer move to measured value




measure r.m.s value

electradynamometer

$L_3$ is free




~~when $L_3$ moves, EMF~~
when current flow $L_1$ and $L_2$,
$L_3$ moves, and current
generate in load

measure $I_{av}^2$

Iron-vane
consists of coil, movable vane, fixed vane
~~when~~ and spring

the measured current flows through coil
which produces magnetic field proportional
to the magnitude of current